From Answer Selection to Differential Reasoning: An Agentic Neuro-Symbolic Framework for Adaptive Evidence Acquisition in Medical Question Answering

Working draft — Paper 3 (MedQA2026). [TBD-EXP#] items await agent experiments; static baselines & seed bounds are verified.

June 2026

Contents

Title (working; alternates below)

From Answer Selection to Differential Reasoning: An Agentic Neuro-Symbolic Framework for Adaptive Evidence Acquisition in Medical Question Answering

Alternates: 1. *Agentic Neuro-Symbolic Differential Diagnosis for Medical Question Answering.* (agentic neuro-symbolic) 2. *Beyond Single-Pass Answers: Differential Reasoning over MedQA Vignettes.* (differential diagnosis) 3. *Adaptive Evidence Acquisition for Medical Reasoning: Selecting What to Check Next.* (adaptive evidence)

4. *Executable Diagnostic Reasoning: UMLS-Grounded Evidence Functions and Belief Updates for MedQA.* (executable reasoning) 5. *Does Adaptive Evidence Selection Help? An Agentic Neuro-Symbolic Study on MedQA.* (MedQA-specific) 6. *Category-Aware Neuro-Symbolic Agents for Clinical Reasoning: A Controlled MedQA Evaluation.* (biomedical informatics) 7. *Belief, Evidence, and Verification: Neuro-Symbolic Agents for Structured Medical Reasoning.* (neuro-symbolic AI)

Abstract

Medical question answering is usually evaluated as static, single-pass answer selection: a model reads a vignette and emits a letter. Clinical reasoning is better described as *differential reasoning* — forming candidate hypotheses, deciding what evidence to examine next, weighing supportive against contradictory findings, updating belief, and producing an auditable trace. Building on a category-balanced MedQA benchmark (Paper 1) and UMLS-grounded verification (Paper 2), we propose and evaluate an **agentic neuro-symbolic framework** that reformulates each question as adaptive evidence acquisition. The system parses the vignette, classifies the reasoning category, generates typed candidate hypotheses, grounds findings and candidates to UMLS, and then runs a controller that maintains a belief state over candidates and **selects evidence functions** — demographic plausibility, symptom/lab support, pathophysiology and treatment-relation checks, contradiction audits, and others — by expected utility under a budget, with redundancy-aware selection and a verification pass, before committing to a final answer and evidence trace. We evaluate on a balanced set of 1,030 MedQA-style questions across five reasoning categories, comparing static zero-shot and chain-of-thought prompting, self-consistency, UMLS symbolic scoring, Chain-of-Verification, and ablated and full agent variants. We report accuracy, category-level performance, evidence efficiency, grounding and contradiction metrics, belief-margin calibration, and error prediction, with bootstrap intervals and paired tests. Analysis of the base models shows roughly **12.6 points of headroom** between the best single model (73.4%) and an oracle that solves any question some model solves (86.0%), and that inter-model agreement strongly tracks correctness — motivating belief-guided adaptive reasoning, while a control analysis shows the headroom is **not** reachable by routing among models. We characterize where adaptive evidence acquisition helps, where it only improves interpretability, and the failure modes that arise when LLM planning meets symbolic grounding. This is a controlled MedQA-style study; we make no clinical-deployment claims.

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1. Introduction

P1 — Answer selection vs reasoning. LLMs score well on MedQA, but emitting a letter says nothing about the reasoning behind it. Treating medical QA as single-pass classification discards the structure of clinical reasoning.

P2 — Clinical reasoning is differential. Clinicians entertain competing hypotheses, decide which evidence to seek next, and revise belief as findings accumulate —

a coarse-to-fine, evidence-guided process producing an interpretable trace.

P3 — MedQA is heterogeneous. A single benchmark mixes Clinical Findings, Diagnosis, Mechanism, Workup, and Treatment — reasoning tasks with different evidence needs, ill-served by one static prompt.

P4 — Paper 1. Base-model behavior is category- and temperature-dependent, with substantial run-to-run instability — accuracy alone hides this structure.

P5 — Paper 2. UMLS grounding and symbolic verification help audit reasoning but static symbolic scoring is brittle (the concept-mismatch problem); symbolic signal is more useful for *verification* than for solving.

P6 — The need for adaptivity. These motivate a system that *adapts* what evidence it examines to the case and its current uncertainty, rather than applying a fixed pipeline. Our seed analysis sharpens the target: an oracle over five base models reaches 86.0% vs the best single model's 73.4% (≈12.6 pp headroom), and inter-model agreement predicts correctness (83.5% when all agree, 23.9% when split) — yet that headroom is **not** capturable by choosing which model to ask (the best model wins every category), so the lever must be *reasoning*, not model routing.

P7 — Proposed framework. We present an agentic neuro-symbolic system (Figure 1, Algorithm 1): parse → categorize → generate candidates → ground → adaptively select evidence functions → update belief → verify → decide, emitting an evidence trace.

P8 — Experiments. We compare static prompting, self-consistency, symbolic-only, Chain-of-Verification, and ablated/full agents across categories, measuring accuracy, efficiency, grounding/verification quality, calibration, and failure modes.

P9 — Contributions (Section 1.1).

1.1 Contributions

1. A reformulation of MedQA as **adaptive evidence acquisition**, with a concrete agentic neuro-symbolic framework and algorithm.
2. An **evidence-function library** of UMLS- and LLM-based clinical checks, with a controller that selects them by uncertainty and expected utility under a budget.
3. **Transparent neuro-symbolic belief updates** and a verification/contradiction auditor that yield ranked candidates, a confidence proxy, and an interpretable evidence trace.
4. A **controlled comparison** against static and symbolic baselines with proper statistics, including the honest finding that base-model headroom is not reachable by model routing.
5. A **failure-mode taxonomy** for agentic neuro-symbolic medical reasoning.
6. Released, reproducible code and analysis. No clinical-deployment claims.

2. Related Work

Clusters, what to cite, positioning, gap. **Medical QA / MedQA** (Jin et al. 2021) — we move from answer selection toward differential reasoning. **Clinical reasoning & differential diagnosis** (cognitive models; Bayesian diagnosis) — we operationalize it for exam QA. **LLMs for medical reasoning** (Med-PaLM, GPT-4 medical) — we add structure and auditability. **Agentic LLMs / tool use** (ReAct, Toolformer) — we instantiate medical evidence functions as tools with symbolic grounding. **Neuro-symbolic AI** — a medical control-loop instantiation. **Biomedical KGs / UMLS** (Bodenreider 2004) — grounding/verification substrate. **CoVe & self-refinement** (Dhuliawala et al.) — we ground verification in a KG and add adaptive selection. **Adaptive computation / active feature selection** (mRMR; active testing) — source of redundancy-aware evidence selection. **Probabilistic reasoning / belief update**, and **explanation faithfulness / clinical interpretability**. *Gap:* no prior work couples category-aware adaptive evidence selection, UMLS grounding, transparent belief updates, and verification into a single controlled MedQA evaluation with a failure-mode analysis.

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3. Background and Motivation

We adapt a coarse-to-fine diagnostic framework (originally for human-motion diagnosis) to clinical-vignette reasoning. The source method estimates coarse context, narrows hypotheses, selectively evaluates biomarkers, maintains belief over competing diagnoses, accumulates probabilistic evidence with minimum-redundancy/maximum-relevance selection, and emits an interpretable trace. **Table A (mapping):**

Motion-diagnosis concept	MedQA analogue
video segment	clinical vignette
subject/body-region visibility	available evidence type (present vs missing)
motion biomarker	clinical finding or relation
gait asymmetry	asymmetric symptom pattern / focal deficit
tremor marker	lab abnormality / symptom pattern / mechanism cue
latent neurological factor	candidate diagnosis / mechanism / workup / treatment
diagnostic graph	UMLS-grounded clinical reasoning graph
adaptive biomarker selection	adaptive evidence-function selection
probabilistic evidence accumulation	answer-choice belief update
mRMR feature selection	redundancy-aware evidence selection
clip diagnostic score	candidate answer posterior
interpretable evidence trace	structured rationale with support/contradiction trace

The translation is the conceptual backbone of the framework: diagnosis as adaptive, evidence-guided belief refinement rather than flat classification.

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4. Dataset

The focused balanced MedQA subset: **1,030** questions, **206 per category**, 4-option (A–D), random baseline 25% (shared with Papers 1–2; **Table 1**). We reuse the verified static baselines and the 144-question hard-core set (all five base models wrong) as a stress target. In MedQA the candidate set is supplied by the answer choices, a simplification relative to open-ended differential diagnosis (Limitations).

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5. Agentic Neuro-Symbolic Framework

Figure 1 shows modules A–J. **A. Case parser** — extract demographics, symptoms, signs, labs, imaging, medications, risk factors, timing, negations, and which evidence is present vs missing. **B. Reasoning-category classifier** — assign category (optionally multi-label) to select a reasoning policy. **C. Candidate hypothesis generator** — options → typed candidates (diseases / mechanisms / interventions / next steps / expected findings). **D. UMLS grounding** — stem findings and candidates → CUIs, semantic types, relation candidates, confidence; preserve negation; flag unmapped concepts (Paper 2). **E. Evidence-function library** — sixteen callable checks returning per-candidate support/contradiction (Table 2; EXP3). **F. Agent controller/planner** — maintains belief, selects the next function by uncertainty × expected utility − redundancy, stops on confidence/budget. **G. Belief-state update** — transparent neuro-symbolic scoring (below). **H. Redundancy-aware selection** — mRMR-style, prioritizing evidence that separates top candidates. **I. Verification/contradiction auditor** — five verifiers → warning flags; optional gated revision. **J. Final decision** — answer, ranked candidates, confidence proxy, support/contradiction table, evidence trace.

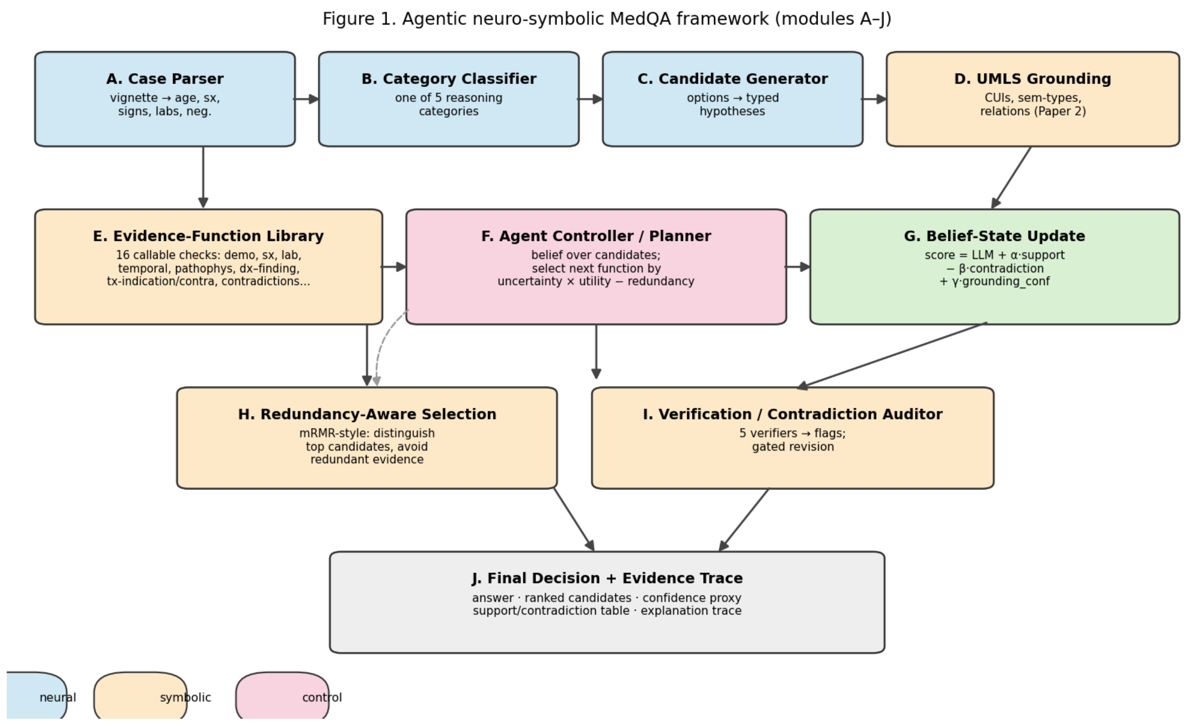

Figure 1. Agentic neuro-symbolic MedQA framework (modules A–J).

5.1 Belief-update formulations (EXP5)

We compare: (A) additive `score += support − contradiction`; (B, recommended) weighted neuro-symbolic `score = LLM + α·support − β·contradiction + γ·grounding_conf`; (C) Bayesian-style prior × evidence likelihood; (D) learning-to-rank over evidence features (future-leaning); (E) calibration-aware confidence from margin, coverage, and flags.

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6. Algorithm

Algorithm 1: Agentic Neuro-Symbolic Medical Reasoning

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Input : vignette x, answer choices A, evidence budget B, UMLS graph K
Output: final answer a*, ranked candidates, evidence trace T
c ← ParseCase(x)                              # module A
cat ← ClassifyCategory(c)                     # module B
H ← GenerateCandidates(A, cat)                # module C
G ← GroundConcepts(c, H, K)                   # module D
belief ← InitPrior(H, G, cat)                 # module G (init)
T ← [ ]
for t = 1 … B do
    u ← Uncertainty(belief)                   # margin / entropy
  f ← SelectFunction(cat, belief, G, T) # module F/H: argmax utility − λ·redundancy
    e ← Execute(f, c, H, G)                   # module E
```

```
    belief ← UpdateBelief(belief, e)      # module G
    T.append((f, e, snapshot(belief)))
    if Margin(belief) ≥ τ or u ≤ ε then break   # adaptive stopping
end for
flags ← Verify(belief, T, G)              # module I
belief ← AdjustOrFlag(belief, flags)      # gated revision
a* ← argmax belief
return a*, Rank(belief), T
```

Figure 3. Agent control loop (Algorithm 1)

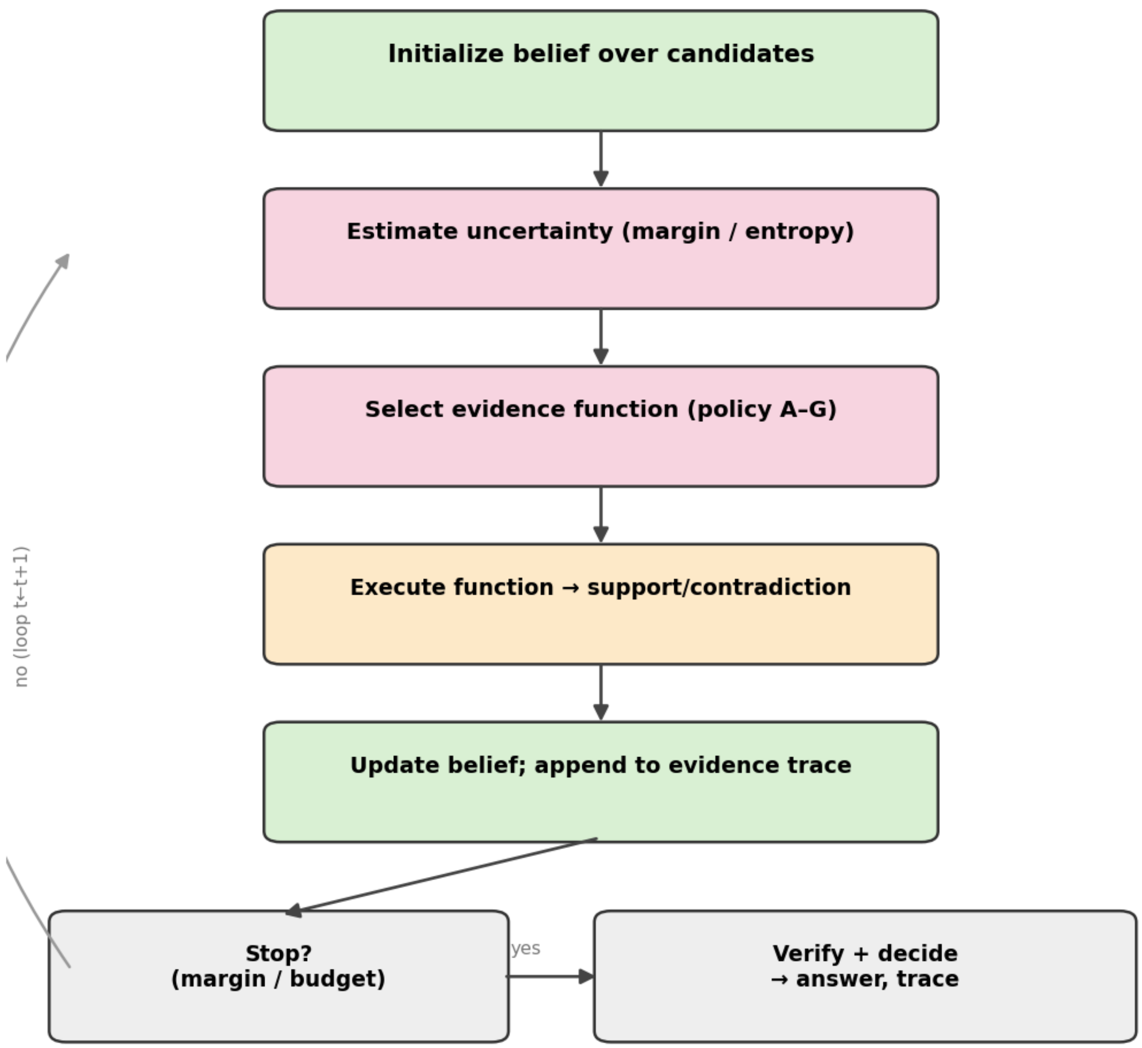

Figure 3. Agent control loop (Algorithm 1): estimate uncertainty, select an evidence function, execute, update belief, stop adaptively, verify, decide.

6.1 Scoring formulation

For candidate `i` and accumulated evidence `E`: - support `S_i = Σ_e w(e)·support(e, i)`; contradiction `C_i = Σ_e w(e)·contra(e, i)` - grounding confidence `g_i ∈ [0,1]`; redundancy penalty `R(f) = sim(f, prior evidence)` - belief `b_i ∝ exp(LLM_i + α·S_i − β·C_i + γ·g_i)` (softmax over candidates) - next-function utility `U(f) = E[Δmargin | f] − λ·R(f)`; stop when `margin ≥ τ` or budget spent - final score = `b_i`, downweighted by verifier flags. α, β, γ, λ, τ tuned on a dev split (EXP5/EXP9); we report sensitivity.

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7. Experimental Design

Conditions (Table 3; EXP8). Baselines: random (25%); static zero-shot LLM (verified Paper 1); static CoT; self-consistency; category-prompt-only; UMLS symbolic-only (Paper 2); UMLS-grounded prompt (Paper 2); CoVe (Paper 2). Agentic: agent w/o UMLS; +grounding no belief-update; +belief no auditor; +auditor no adaptive selection; **full agent**; full agent fixed-budget; full agent dynamic-stopping; oracle bound (86.0%). **Statistics:** bootstrap CIs, McNemar, paired bootstrap for category deltas, logistic regression for correctness prediction, AUROC/AUPRC for error prediction, mixed-effects for method×category, correlation of budget vs accuracy, ablation significance with multiple-comparison correction — reusing the Paper 1/2 harness. **Anti-overclaim:** modest, category-specific, or interpretability-only gains will be reported as such, with CIs, and not headlined as accuracy wins.

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8. Results

Agent numbers are placeholders; static baselines and seed bounds are verified.

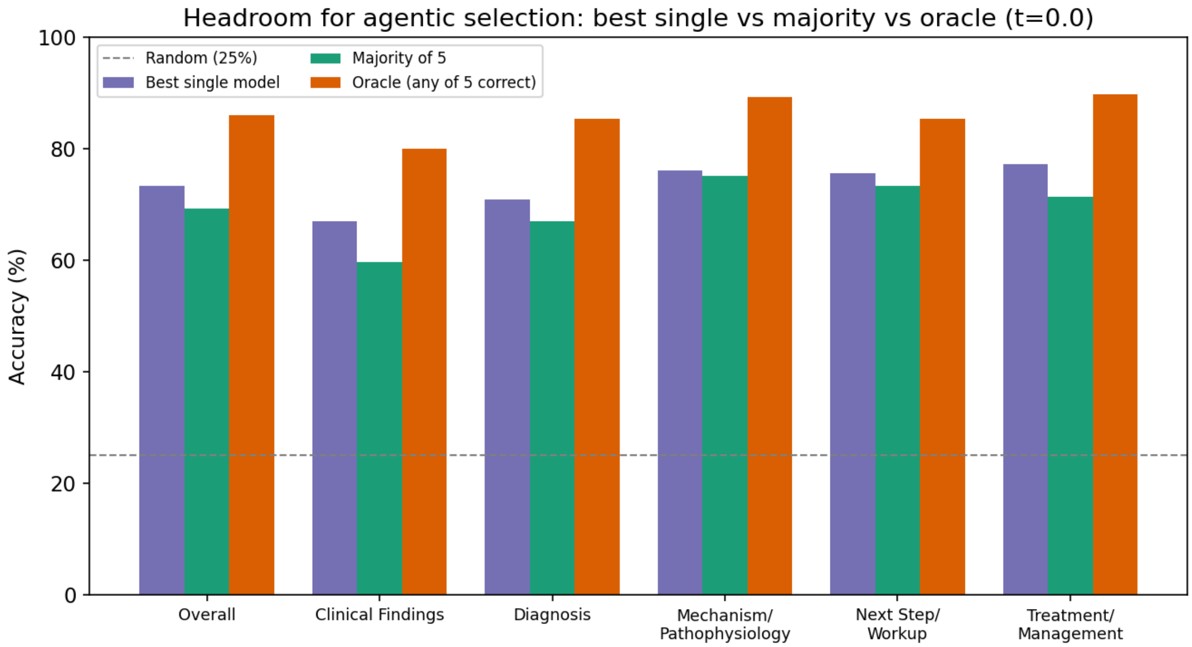
8.1 Overall accuracy by method (Table 4). Static LLM-only baseline: best single Qwen2.5 32B 73.4% (Paper 1). Symbolic-only / UMLS-prompt / CoVe [TBD-Paper2]. Agent variants [TBD-EXP8], positioned against best-single 73.4% and oracle 86.0%.

8.2 Category-specific accuracy (Table 5, Figure 6). [TBD-EXP8] per method × category; test whether agentic gains concentrate in Diagnosis/Workup/Treatment (RQ7). Seed caution: since the best model wins every category, category gains must come from reasoning, not model choice.

8.3 Agentic efficiency (Table 6, Figures 5, 7). Mean evidence calls, stopping depth, function-selection distribution by category, accuracy vs budget (diminishing returns), adaptive vs fixed-policy delta [TBD-EXP4/EXP6/EXP9].

8.4 Belief calibration (Figure 4). Belief-margin reliability and the margin-vs-correctness curve [TBD-EXP5], compared to the verified agreement-vs-correctness seed (5-agree 83.5% → 2-agree 23.9%).

8.5 Grounding & verification (Table 8, Figure 8). Grounding coverage, contradiction scores, verifier precision/recall, and error-prediction AUROC [TBD-EXP7].



Seed figure. Headroom for agentic selection: best single model vs majority-of-5 vs oracle (any-of-5 correct) at t=0.0. Oracle exceeds best single by ~10–15 pp; majority voting is below best single.

9. Ablation Studies

[TBD-EXP9] Remove, one at a time: UMLS grounding, category classifier, adaptive selection (→ fixed sequence), belief updates, contradiction auditor, redundancy penalty, semantic-type compatibility, pathophysiology checker, contraindication checker; plus LLM-only vs symbolic-only evidence functions, random selection, no-adaptivity, no-final-verification, no-revision. Report accuracy deltas with CIs, rank component contributions, and flag any component that does not help (a legitimate negative result given the §8.2 caution). Evidence-budget sweep B=1..N (Figure 10).

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10. Case Studies

[TBD-EXP10] Five worked traces: (1) Diagnosis improved by adaptive evidence selection; (2) Treatment saved by a contraindication check; (3) Workup helped by next-best-test reasoning; (4) Mechanism helped by pathophysiology-chain reasoning; (5) a failure where the agent selects misleading evidence or grounding fails. Each shows vignette, choices, evidence calls, belief trajectory (Figure 4), flags, and final answer, drawn where possible from the 144 hard-core items. Figure 9 shows an example evidence trace.

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11. Discussion

The thesis is methodological: medical QA is more faithfully modeled as adaptive differential reasoning than as single-pass selection. The research question is not only whether the agent raises accuracy, but whether adaptive evidence selection, grounding, contradiction auditing, and belief updates yield more **reliable, interpretable, uncertainty-aware** reasoning. The seed analysis frames expectations honestly: there is real headroom (oracle 86.0% vs 73.4%), but it is invisible to model routing and not reachable by naive voting, so any gain must come from *better reasoning within a model*. We expect the agent's clearest value in **error detection and interpretability** (belief margin and contradiction flags predicting wrong answers) and in specific categories, with accuracy gains possibly modest — and we will report that plainly.

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12. Limitations

MedQA is exam-style, not clinical practice; multiple-choice supplies candidates, simplifying the differential. Agentic tool calls add compute. UMLS is incomplete and not a causal reasoning engine; concept mismatch (Paper 2) remains unresolved. LLM rationales may be unfaithful, so trace-quality needs manual validation. Belief weights (α, β, γ, λ, τ) require tuning and may overfit dataset structure; evidence-function outputs can be noisy. The agent is **not yet implemented** in this snapshot — all agentic results are pending experiments. No patient-safety or clinical-deployment claims are made.

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13. Future Work

Open-ended differential diagnosis without supplied choices; retrieval-augmented literature grounding; integration with SNOMED CT, RxNorm, MeSH, and guidelines; **learned** evidence-selection policies; clinician evaluation of evidence traces; application to real clinical notes; uncertainty-aware abstention; causal medical knowledge graphs; multi-agent specialist reasoning; and dedicated patient-safety evaluation. The framework's grounding, verification, and belief machinery are the substrate for these extensions.

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14. Conclusion

We reformulate MedQA as adaptive evidence acquisition and present an agentic neuro-symbolic framework — parse, categorize, hypothesize, ground, adaptively gather evidence, update belief, verify, and decide with an interpretable trace. Grounded in verified base-model analysis (≈12.6 pp oracle headroom, agreement-tracks-correctness, no model-routing gain), the study asks where adaptive, grounded, verified reasoning improves reliability and interpretability over static prompting, and characterizes its failure modes. Code and analysis are released; we make no clinical-utility claims.

Appendix / Reproducibility (outline)

- A. Prompt templates (parser, classifier, planner) — EXP1/EXP4.
- B. Evidence-function specifications and unit tests — EXP3.
- C. Belief-update equations and tuned weights — EXP5.
- D. Full per-condition, per-category tables — EXP8/EXP9.
- E. Agent trace schema; example traces — EXP4/EXP10.
- F. Failure-mode annotation guide + κ — EXP10.

Figures and tables (asset map)

- Table 1 dataset · Table 2 evidence-function library (EXP3) · Table 3 system variants · Table 4 overall accuracy (EXP8) · Table 5 category accuracy (EXP8) · Table 6 agentic efficiency (EXP4/6) · Table 7 ablations (EXP9) · Table 8 grounding/verification (EXP7) · Table 9 failure taxonomy (EXP10).
- Figure 1 framework (`figures/fig1_framework.png`, ready) · Figure 2 vignette→trace mapping [TBD-EXP4] · Figure 3 control loop (`figures/fig3_control_loop.png`, ready) · Figure 4 belief update example [TBD-EXP5] · Figure 5 accuracy vs budget [TBD-EXP9] · Figure 6 category-gain heatmap [TBD-EXP8] · Figure 7 function-selection distribution [TBD-EXP4] · Figure 8 contradiction vs error prob [TBD-EXP7] · Figure 9 example evidence trace [TBD-EXP10] · Figure 10 ablation chart [TBD-EXP9]. Seed figure: oracle headroom (`figures/fig_seed_oracle_headroom.png`).

Publication strategy (notes)

arXiv-first. Venues: ML4H, CHIL, AMIA (informatics), neuro-symbolic / KR workshops, medical-AI workshops. **Minimum workshop package:** EXP1–EXP4 + EXP8 (a working agent vs static prompting on some categories). **Full conference:** add EXP5–EXP7 and EXP9 (belief, verification, ablations, budget). **Journal:** add EXP10 with clinician trace review and broader baselines.